Web-CogReasoner: Towards Multimodal Knowledge-Induced Cognitive Reasoning for Web Agents

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Paper version: arXiv 2508.01858v3 · CUA Radar Reel poster, not made or endorsed by the authors

Problem

> *web agents must first acquire sufficient knowledge to effectively engage in cognitive reasoning*

Knowledge enhancement is **ad hoc**; Bloom's taxonomy — Factual, Conceptual, Procedural — supplies the curriculum.

Motivation

1. **Ad hoc.** Knowledge enhancement lacks systematic grounding.
2. **Bottleneck.** Weak on specialized web tasks.
3. **Flat SFT.** Facts, concepts, how-to mixed.

Evaluation setup

WebVoyager · 643 online | Online-Mind2Web · live | Web-CogBench · 876 | Qwen2.5-VL-7B base | SFT · 8×A800 80GB | seeds not reported

- **Online:** WebVoyager (643 queries) + Online-Mind2Web (cross-task / cross-web).
- **Offline:** own Web-CogBench (876) + VisualWebBench; UI-TARs-7B off online (no scripts).

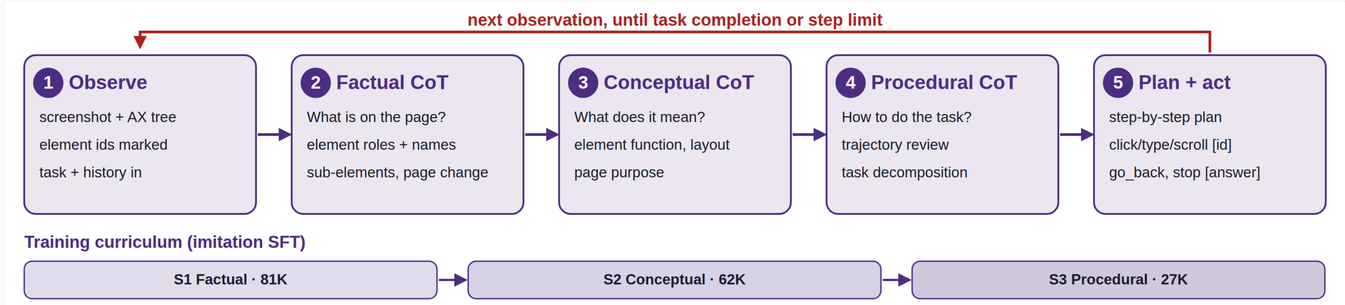
Our read (CUA Radar)

62	★★	transfer
score of 100	band	evidence level

- **Reports:** live online eval; unseen websites; ablation; judge check; code.
- **Not reported:** seeds; budget match. **Partial:** UI-TARs off online. **Flag:** AP5.
- Pattern C21 (SFT on synthesized reasoning). Authors' numbers, not rerun.

Method: a knowledge curriculum + knowledge-driven CoT

- **Sees / acts:** screenshot + AX tree in; click / type / scroll / stop [answer] out, on element ids.
- **Curriculum:** SFT of Qwen2.5-VL-7B on S1 Factual → S2 Conceptual → S3 Procedural (81K / 62K / 27K).
- **KCoT:** layout, key-element analysis, intent, then plan → action, grounded in the three layers.

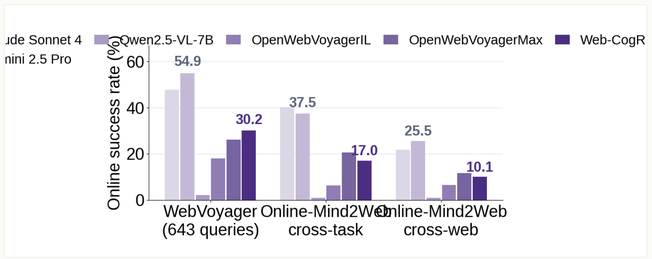


Our redraw (not the paper's figure): the per-step KCoT loop, with the 3-stage training curriculum below.

$$P = (S, A, O, K, T, R)$$

web interaction as a POMDP with internal knowledge K

Key results: online success rate (%)



Tables 4-5, redrawn: live web, single runs (no seeds reported); 7B SFT vs commercial LVMs.

Open-source best 30.2% on WebVoyager; Gemini 2.5 Pro leads at 54.9%

SO WHAT → *Online-Mind2Web: ours 17.0 / 10.1 vs OpenWebVoyagerMax 20.5 / 11.7 (retrained on high-error sites).*

Ablation: WebVoyager online success (%)

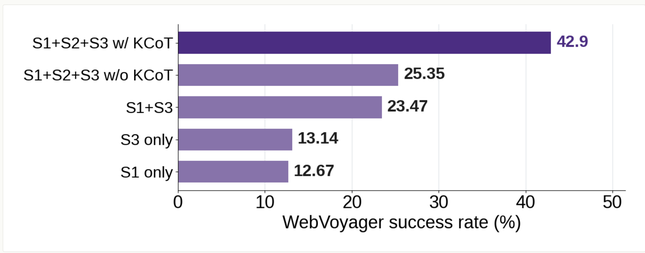


Table 9, redrawn: 4-site subset (Amazon, Cambridge, Coursera, GitHub).

SO WHAT → *KCoT activates the knowledge: 42.9% → 25.35% without it; factual S1 nearly doubles S3-only (23.47 vs 13.14). On Web-CogBench the cumulative curriculum lifts overall 69.8 → 72.1 → 78.3 → 84.4, while single-stage SFT underperforms (S2 only 61.96, S3 only 60.66, vs 84.44 for the full stack): the layers are hierarchically dependent. Curriculum also beats mixed multi-task SFT, which stops exploring early (Fig. 7).*

Headline Numbers

30.2%
WebVoyager success
best open-source agent in the paper's tables; Gemini 2.5 Pro 54.9%

- **17.0%** Mind2Web cross-task
- **84.4** Web-CogBench (offline)
- **7.00** avg steps / success

The paper's own controls

CONTROL	OURS	CONTEXT
Base model online	30.2%	Qwen base: 2.2%
Unseen sites	10.1%	OWV-Max 11.7%
w/o KCoT	25.35%	42.9% with KCoT
Judge agreement	96.0-98.7%	3 LVM judges, ±1 pt

Takeaway

Knowledge-curriculum SFT + KCoT gives the best open-source WebVoyager score in these tables (30.2%); Gemini 2.5 Pro stays ahead (54.9%), and cross-web transfer trails the retrained OpenWebVoyagerMax (10.1% vs 11.7%).

14 sites crawled | 3 knowledge layers | 81K/62K/27K SFT | no seeds reported

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